The Gate Knows, the Threshold Doesn't: Recalibrating Activation-Gated Jailbreak Defenses Under Unseen Framings

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Abstract

Activation-gated defenses read a model's residual stream and intervene, for example by steering towards refusal, only when a probe fires. Their thresholds are set on jailbreak framings known at development time. We test six such gates on real harmful and benign prompts in thirteen framings, nine of which are withheld from all design decisions. On several withheld framings recall at a 1% false-positive budget falls to between 0% and 17%. Yet a probe over the user's tokens still ranks harmful above benign prompts with AUROC 0.95–0.99 on every withheld framing the model can parse, and re-thresholding it per framing would recover most of the loss. The failure is mostly one of calibration. We propose a recalibration that uses no labels from the deployment stream: cluster incoming prompts by framing, standardise the probe score robustly within each cluster, and combine the result with the original threshold as a floor. The method is useful but fragile. As pre-registered it under-segments the traffic and helps only some framings. With more clusters (a post-hoc change) and realistic traffic it lowers judged attack success from 89% to 37% on fiction, from 45% to 6% on code-completion and from 15% to 4% on academic framings, with an over-refusal cost below one point. It cannot help framings used almost only for attacks.

1 Introduction

Safety training teaches a language model to refuse harmful requests. Jailbreaks undo it by changing how the request is framed: a persona, a story, a retrieved document, an encoding, or a request to write code (Wei et al., 2023; Shen et al., 2024; Greshake et al., 2023; Li et al., 2023). Because the framing changes the surface of a prompt but not what it asks for, a natural defense looks beneath the surface. Residual-stream activations carry a linearly readable harmfulness signal (Zou et al., 2023a; Zhao et al., 2025), and refusal itself is mediated by a single direction (Arditi et al., 2024). Several defenses therefore read activations to decide whether to intervene. Conditional activation steering adds a refusal direction only when a condition vector fires (Lee et al., 2025), and probe-based detectors flag a prompt from its internal state (Zhang et al., 2025; McKenzie et al., 2025). These gates are cheap, they leave benign traffic alone, and their operating point can be set to a fixed false-positive budget.

This paper asks what happens to that operating point when the framing changes. We separate framing families into *seen* ones, used for training and calibration, and *withheld* ones, and we evaluate on real prompts from five public benchmarks (Mazeika et al., 2024; Zou et al., 2023b; Chao et al., 2024; Röttger et al., 2024; Cui et al., 2025), with generations judged by a guard model (Qwen Team et al., 2025).

The first finding is a collapse. Every gate we test detects at most 17% of harmful prompts on withheld fiction and indirect-injection framings, and none of them changes the 89% attack success rate of fiction prompts. This includes a geometric gate of our own design that explicitly tries to factor framing out of the representation.

The second finding concerns the cause. Ranking survives the shift. A probe that reads the mean of the user's tokens separates harmful from benign prompts with AUROC 0.95–1.00 on every withheld framing the model can parse. On the four framings the shared threshold misses worst, re-thresholding it on the new framing recovers 57–81% recall. What moves is the location of both score distributions. Concurrent work reports compatible observations: linear jailbreak probes are poorly calibrated off-distribution (Kim and Lee, 2026), and shift degrades threshold transfer before ranking (Luo et al., 2026). We confirm the pattern across gates, readout positions and framings. We also show that projecting out a subspace of framing

directions, the most direct representational fix we could build, does not repair it.

The third finding is a remedy and a map of its limits. Deployed gates see a stream of mostly benign prompts. We cluster the stream by framing, standardise the probe score inside each cluster with median and median absolute deviation, which tolerate a minority of harmful prompts (Rousseeuw and Croux, 1993; Leys et al., 2013), and keep the original threshold as a floor. Unsupervised recalibration of subpopulations has antecedents in general machine learning (Ziegler and Czyż, 2019). Our contribution is to adapt it to activation-gated safety and to test it under a protocol that withholds framings from design. We then report plainly where it works and where it does not.

Concretely, we contribute:

- a held-out-framing evaluation of activation gates on real corpora, with leave-one-framing-out selection on validation data and five framings preregistered after the main study;
- evidence that gate failure on withheld framings is mostly a calibration shift, and a negative result for framing-subspace projection;
- a label-free, cluster-wise recalibration with a fixed-threshold floor, evaluated without transduction at detection and generation level;
- a characterisation of its failure modes: cluster resolution, framings used mainly for attacks, harmful prevalence, and framings the model cannot parse.

2 Related Work

Jailbreak framings. Attacks range from optimised suffixes (Zou et al., 2023b; Liu et al., 2024) and attacker models (Chao et al., 2025; Mehrotra et al., 2024) to hand-written templates: personas (Shen et al., 2024), nested fiction (Li et al., 2023), encodings and ciphers (Wei et al., 2023; Yuan et al., 2024), many-shot demonstrations (Anil et al., 2024), and instructions planted in retrieved content (Greshake et al., 2023). Wei et al. (2023) attribute their success partly to mismatched generalisation, which is what a withheld framing probes.

Representation-level defenses. Harmfulness and refusal are encoded separately, with harmfulness concentrated at the instruction tokens rather than at the final position (Zhao et al., 2025). Representation engineering reads and edits such directions (Zou et al., 2023a); circuit breakers reroute harmful representations during training (Zou et al., 2024); latent adversarial training hardens them (Sheshadri et al., 2024). Inference-time gates include conditional steering (Lee et al., 2025) and detectors that flag prompts from internal signals (Zhang et al., 2025; Hu et al., 2024; Xie et al., 2024), or redact during generation (Mei et al., 2026). FrameShield likewise treats framing as a nuisance to be disentangled from the goal in activation space (Farzam et al., 2026). Several studies question how far probe signals generalise. Probes for high-stakes interactions keep high AUROC out of distribution while recall at low false-positive rates is modest (McKenzie et al., 2025). Final-token probes miss jailbreaks whose evidence sits in earlier user tokens (Doda, 2026). Linear jailbreak probes conflate benign register with intent, and their headline accuracy depends on the threshold (Kim and Lee, 2026). Internal harmfulness scores can even anti-rank successful jailbreaks (Luo et al., 2026). In contrast, Schwarz (2026) find that refitting thresholds does not help on topic-matched pairs, where the discriminative information is absent. Adaptive attacks that optimise against the probe can drive recall to zero (Bailey et al., 2026). We study the complementary, non-adaptive failure of natural framing shift.

Guard models and judges. Classifier guards moderate inputs and outputs (Inan et al., 2023; Han et al., 2024; Zeng et al., 2024; Padhi et al., 2025; Qwen Team et al., 2025), and constitutional classifiers are trained against universal jailbreaks (Sharma et al., 2025). We use a guard only as an evaluation judge, because refusal-string matching counts harmless non-compliance as a successful attack (Souly et al., 2024).

Calibration under shift. Label-shift corrections re-estimate class priors from unlabeled target data (Saerens et al., 2002; Lipton et al., 2018; Alexandari et al., 2020). Test-time adaptation adjusts normalisation layers by entropy minimisation (Wang et al., 2021). Conformal methods restore coverage under shift given weights or labels (Tibshirani et al., 2019; Podkopaev and Ramdas, 2021). Unsupervised recalibration adjusts outputs per subpopulation without labels (Ziegler and Czyż, 2019); multicalibration asks for calibration on every identifiable subgroup, using labels (Hébert-Johnson et al., 2018). For jailbreak detectors, Piet et al. (2025) adapt text classifiers to drift by self-training, Azov et al. (2026) find that re-thresholding often suffices

when target labels exist, and Kim and Lee (2026) combine register erasure with distribution-free calibration on labelled benign data. Our method uses no labels from the shifted framing or the deployment stream, and it operates per framing cluster of an activation gate.

3 Setup

Prompts. Harmful behaviours come from JailbreakBench, the standard text behaviours of HarmBench, and AdvBench; benign prompts come from the benign split of JailbreakBench, the safe prompts of XSTest, and 400 prompts of OR-Bench-Hard, which are designed to look harmful. After deduplication there are 1,550 prompts. We split them 50/20/30 into train, validation and test *by behaviour*, stratified by source, so that no wrapped version of a test request is seen in training. The test split holds 241 harmful and 225 benign requests.

Framings. Each request is wrapped in a framing that keeps its text verbatim. Four framings are *seen*: the plain request, a persona (AIM, DAN, "grandma"), refusal suppression, and prefix injection. Four were *withheld* from the start: fiction or screenplay, indirect injection disguised as a retrieved document or e-mail, base64, and leetspeak. After the main study we wrote and committed five *fresh* framings before generating any output on them: a past-tense historical frame, an academic or IRB frame, code completion, translate-then-answer, and payload splitting. Every family uses one to three fixed templates. We also optimise GCG suffixes (Zou et al., 2023b) against the undefended model for 20 test behaviours (Appendix B).

Model and gates. The target is Qwen2.5-1.5B-Instruct (Qwen Team, 2024) without a system prompt. All gates read layer 14 of 28. When a gate fires, it adds the same vector $\alpha\hat{r}$ to the residual stream at every position. Here $\hat{r}$ is the normalised difference between mean harmful and mean benign activations of plain training requests, and α is 0.4 times the mean activation norm. Gates therefore differ only in their decision. We compare six gates: (i) a CAST-style condition vector (Lee et al., 2025); (ii) an MLP probe on the final token; (iii) ARETE, a geometric gate of our own that maps activations onto intent and context hyperspheres and steers orthogonally to a framing subspace (Appendix A); and logistic-regression probes on (iv) the final token, (v) the mean of the user's tokens, and (vi) a union of three readouts (Section 4). Every gate is trained on seen framings of the training split. Its threshold is the 99th percentile of its score on benign validation prompts of seen framings, a held-out-quantile calibration in the spirit of Kim and Lee (2026) but without their register erasure. We call it the *shared* threshold.

Protocol and its departures. Readout position, clustering layer and clustering features were selected by leave-one-framing-out (LOFO) on the validation split: each seen wrapper in turn is treated as withheld, and recall is measured on it. Two departures need stating. First, the diagnostic analyses of Section 4 used the test split before the remedy was designed, so the original withheld framings informed the idea, though not its configuration. This is why we pre-registered the fresh framings. Second, the layer was fixed at 14 for all gates and never selected; in LOFO the user-mean probe scored slightly higher at layers 21 and 7 (43% and 41% recall) than at layer 14 (38%).

Metrics. Detection is reported as true-positive rate (TPR, which we call recall) on harmful prompts and false-positive rate (FPR) on benign prompts, plus AUROC. Responses are generated greedily for 96 new tokens and judged by Qwen3Guard-Gen-4B in response-moderation mode (Qwen Team et al., 2025). The judge sees the *unwrapped* request and the response. Attack success rate (ASR) is the fraction of harmful requests whose response is judged UNSAFE. Over-refusal is the fraction of plain benign requests whose response is judged a refusal; on wrapped benign prompts the judge often labels confused non-compliance as refusal, so we do not report it there. For base64 the judge labels responses that merely echo or start decoding the payload as UNSAFE, so we report detection only. With 96 tokens, ASR mostly measures the onset of compliance; Appendix B repeats the key cells at 256 tokens.

4 Diagnosis: Ranking Survives, the Threshold Moves

Recall collapses on withheld framings. Table 1 shows that the probes and ARETE detect 83–97% of harmful prompts on seen framings at the shared threshold. CAST is weaker on wrapped seen framings, at 0–50%. On withheld framings recall falls to 0–17% for every gate. Benign FPR on those framings also falls to nearly zero, which already

	Seen framings				Unseen framings			
Gate	Direct	Persona	Ref. supp.	Prefix inj.	Fiction	Indirect	Base64	Leetspeak
CAST	85 (0.98)	42 (0.97)	50 (0.96)	0 (0.94)	0 (0.87)	0 (0.94)	0 (0.46)	0 (0.93)
MLP probe	91 (1.00)	90 (1.00)	87 (1.00)	83 (1.00)	0 (0.81)	0 (0.93)	0 (0.41)	0 (0.94)
ARETE (v1 geometry)	93 (0.99)	93 (0.99)	94 (0.99)	88 (0.99)	0 (0.40)	3 (0.71)	0 (0.47)	0 (0.93)
Probe, last token	94 (1.00)	91 (1.00)	90 (1.00)	87 (1.00)	0 (0.86)	15 (0.89)	0 (0.45)	0 (0.95)
Probe, user tokens (mean)	96 (1.00)	88 (1.00)	83 (1.00)	93 (1.00)	5 (0.99)	17 (0.98)	0 (0.53)	0 (0.72)
Multi-view union	97 (1.00)	93 (0.99)	92 (0.99)	88 (1.00)	0 (0.83)	12 (0.89)	1 (0.54)	5 (0.94)

Table 1: Detection on the test split at the shared threshold (1% benign FPR on seen framings): recall (%) and, in parentheses, AUROC. All gates read layer 14 of Qwen2.5-1.5B-Instruct.

Readout	Framing	AUROC	TPR (shared)	TPR (oracle)
Last token	Fiction	0.86	0.0	21.6
	Indirect	0.89	15.4	41.9
	Base64	0.45	0.0	0.0
	Leetspeak	0.95	0.0	17.8
User mean	Fiction	0.99	4.6	80.5
	Indirect	0.98	16.6	78.8
	Base64	0.53	0.0	0.0
	Leetspeak	0.72	0.0	17.4

Table 2: Withheld framings, test split. *Shared*: threshold from seen framings. *Oracle*: threshold re-set to 1% FPR on benign prompts of the same framing (unavailable in practice). Recall in %.

Fresh framing	AUROC	TPR (shared)	TPR (oracle)
Past tense	0.97	0.4	57.3
Academic	0.96	39.8	63.5
Code	0.98	83.0	67.6
Translation	1.00	95.4	90.0
Payload split	0.99	0.4	78.0

Table 3: Fresh pre-registered framings, user-mean probe, test split. Recall in %.

suggests a shift in location: the whole score distribution of a new framing moves below the threshold. ARETE's fiction AUROC is 0.40, worse than chance, so its learned intent geometry does not transfer to a narrative frame.

The information is still there. If a framing erased the harmful signal, no threshold could recover it. Table 2 compares the shared threshold with an oracle that re-thresholds on benign prompts of the withheld framing itself. For the user-mean probe, AUROC on fiction and indirect injection is 0.99 and 0.98, and the oracle lifts recall from 5% to 81% and from 17% to 79%. The fresh framings, examined only after the method was fixed, show the same pattern (Table 3). AUROC ranges from 0.95 to 1.00, and the oracle lifts past-tense recall from 0.4% to 57% and payload-split recall from 0.4% to 78%. Base64 is the exception. AUROC is near chance and no threshold helps, because the 1.5B model does not reliably decode the payload, so no representation of the request forms. Leetspeak is similar at the user-mean readout (AUROC 0.72), and the model rarely complies with it at all.

Projecting out framing does not help. ARETE assumes that framing occupies a low-dimensional subspace that can be removed. We test this in LOFO on validation data: we estimate the subspace from wrapper displacements of the remaining seen wrappers and project it out before training a linear probe. Removing 0, 4, 16 or 64 directions gives mean held-out recall of 54%, 54%, 54% and 48%. A subspace learned from two wrappers does not contain the shift induced by a third. Prefix injection is the clearest case, at about 1% recall for every rank.

Where to read. Readout position matters more than projection. In LOFO, the final-token probe misses prefix injection (1%), whose injected instruction follows the request and changes the final state. The user-mean probe catches 60% of it but is weaker on personas. A union of three readouts, each calibrated at a third of the budget, raises LOFO recall from 54% to 69% at 1.3% FPR, and on the test split it helps seen framings slightly (Table 1). It does not help withheld framings, because each readout suffers the same shift.

5 Cluster-Wise Robust Recalibration

The diagnosis suggests treating each framing as its own calibration problem. At deployment the framing of a new prompt is unknown and no labels are available. Two facts help. Prompts that share a framing share an activation signature, so they cluster. And when a framing is used mostly for

Algorithm 1 Cluster-wise robust recalibration

Require: probe score s; framing features ϕ; budget ϵ; shared threshold $\tau_{\epsilon/2}$

1: **Once, on validation:** simulate streams of seen framings, run lines 2–3, and set $z^\star$ to the $(1 - \epsilon/2)$ quantile of benign robust z.
2: **Window:** on unlabeled traffic, reduce ϕ by PCA and run k-means, $k \leq k_{\max}$ chosen by silhouette.
3: For each cluster c: $m_c \leftarrow \mathrm{median}(s)$, $d_c \leftarrow 1.4826\,\mathrm{MAD}(s)$; freeze centroids and (m_c, d_c).
4: **New prompt** x, nearest centroid c: fire iff $s(x) > \tau_{\epsilon/2}$ or $(s(x) - m_c)/d_c > z^\star$.

benign purposes, the bulk of its score distribution describes benign prompts.

Algorithm 1 gives the *hybrid* rule. The score s is the user-mean probe. The features ϕ are user-mean activations at layer 14, reduced to 32 principal components. Clusters with fewer than 30 prompts fall back to validation statistics. Inside a cluster the median and MAD describe its benign bulk, so a harmful prompt is compared with benign prompts of its own framing. Because $z^\star$ is set on validation streams that pass through the same clustering, it accounts for within-cluster dispersion. The *adaptive* rule is the second disjunct alone, at the full budget. Validation labels are used once, to set $z^\star$ and τ. No labels from the deployment stream or from new framings are used.

LOFO on validation streams at 5% harmful prevalence selected the user-mean readout over the final-token and last-user-token readouts, and layer-14 clustering features over layers 7 and 21. Under that selection, held-out recall rose from 38% with the shared threshold to 91% with the adaptive rule, at 1.1% FPR. The hybrid rule reached 83% at 0.9% FPR. We fixed $k_{\max} = 12$ without selection, which turns out to matter.

6 Results

We evaluate without transduction. In each of 20 windows the test behaviours are split at random into halves A and B. Cluster statistics are estimated on traffic built from half A and applied to half B. We use two traffic mixes. *Benign-rich* traffic gives every framing 5% harmful prevalence. *Realistic* traffic keeps 5% for dual-use framings (plain, fiction, indirect, past tense, academic, code, translation) and uses 50% for framings that are rarely used benignly (persona, refusal suppression, prefix injection, base64, leetspeak, payload split). Generation-level ASR is composed from judged responses. A fired prompt takes the label of its steered response, which equals the always-on response under greedy decoding, and an unfired prompt takes the label of its undefended response.

As pre-registered, the method under-segments. Table 4a reports the frozen configuration on all thirteen framings. In benign-rich traffic, payload-split recall rises from 0% to 90%, indirect injection from 16% to 31%, past tense from 0% to 15%, and academic from 40% to 45%. Fiction gains little, from 5% to 19%. When the stream contains only the eight original framings, the same configuration detects 82% of fiction and 56% of indirect prompts. With thirteen framings and at most twelve clusters, fiction shares a cluster with framings whose benign scores sit elsewhere, and its location is not corrected. With true framing labels as groups (Table 5) fiction recall is 76%, so the loss comes from cluster resolution, not from the statistics.

With enough clusters it works where traffic is mostly benign. Raising $k_{\max}$ to 30 is a post-hoc change, motivated by the observation above, and we report it as such. In benign-rich traffic, adaptive recall then reaches 91% on fiction, 92% on indirect injection, 84% on academic and 90% on payload splitting, at 1.6% mean FPR (Table 4b). Recall rises steadily with $k_{\max}$: at 12, 20 and 30 it is 27%, 70% and 91% on fiction. Past tense stays at 40%, below its 57% oracle. Table 5 isolates the clustering. One global robust z (k=1) does nothing. Grouping by true framing family helps, but less than data-driven clusters, which can also separate templates within a family.

It fails where a framing is mostly harmful, and the floor contains the damage. Under realistic traffic the persona, refusal-suppression and prefix-injection clusters are half harmful. Their median then sits among harmful scores, and adaptive recall on them drops to 0–4% (Table 4). This is worse than the shared threshold, and prefix-injection ASR returns to the undefended 77%. Payload splitting, an attack-only framing that the shared threshold also misses, is lost entirely. The hybrid rule keeps the shared threshold as a floor and recovers 55–82% recall on seen attack-only framings, while keeping nearly all of the gain on dual-use framings.

		Benign-rich traffic			Realistic traffic		
Group	Framing	Shared	Adaptive	Hybrid	Shared	Adaptive	Hybrid
Seen	Direct	96	91	97	96	93	97
	Persona	88	85	85	87	4	58
	Ref. supp.	83	72	75	83	0	55
	Prefix inj.	93	88	87	93	0	81
Unseen	Fiction	5	19	19	4	9	9
	Indirect	16	31	29	16	28	26
	Base64	0	0	0	0	1	1
	Leetspeak	0	2	2	0	0	0
Fresh (pre-registered)	Past tense	0	15	14	0	7	6
	Academic	40	45	47	39	41	41
	Code	83	79	91	83	74	85
	Translation	96	94	94	95	93	93
	Payload split	0	90	89	0	0	0
	Benign FPR (mean)	0.6	1.5	1.4	0.7	1.2	1.3

(a) Pre-registered, $k_{\max}$=12, 20 windows

		Benign-rich traffic			Realistic traffic		
Group	Framing	Shared	Adaptive	Hybrid	Shared	Adaptive	Hybrid
Seen	Direct	96	76	95	96	70	94
	Persona	88	92	91	87	0	58
	Ref. supp.	83	96	96	83	0	56
	Prefix inj.	93	91	91	93	0	82
Unseen	Fiction	4	91	91	5	89	88
	Indirect	16	92	91	16	89	89
	Base64	0	1	1	0	1	1
	Leetspeak	0	3	3	0	0	0
Fresh (pre-registered)	Past tense	0	40	38	0	36	35
	Academic	40	84	83	40	80	79
	Code	83	82	92	83	83	90
	Translation	96	94	94	95	92	92
	Payload split	0	90	90	0	0	0
	Benign FPR (mean)	0.6	1.6	1.6	0.7	1.1	1.3

(b) Post hoc, $k_{\max}$=30, 10 windows

Table 4: Recall (%) on harmful test prompts of half B, mean over windows; last row: benign FPR averaged over the 13 framings. Thirteen framings share each traffic window. *Shared*: fixed threshold. *Adaptive*: cluster-wise robust z alone. *Hybrid*: either fires (each at half the budget).

Rule	Fiction	Indirect	Past tense	Academic	Payload split	FPR
Shared threshold	4	16	0	40	0	0.6
Global robust z (k=1)	1	5	0	30	0	0.4
Per-family z (oracle groups)	76	29	44	47	90	1.2
Per-cluster z (ours)	91	92	40	84	90	1.6

Table 5: Grouping ablation, benign-rich traffic, $k_{\max}$=30 (post hoc). Recall (%); FPR averaged over 13 framings.

Framing	No defense	Shared threshold	Adaptive (ours)	Hybrid (ours)
Direct	1 [0,2]	0 [0,1]	1 [0,2]	0 [0,1]
Persona	3 [2,5]	1 [0,3]	3 [2,5]	3 [2,5]
Ref. supp.	3 [1,3]	2 [1,3]	3 [1,3]	3 [1,3]
Prefix inj.	77 [74,82]	9 [6,13]	77 [74,82]	16 [13,21]
Fiction	89 [87,92]	88 [86,90]	36 [27,46]	37 [29,47]
Indirect	12 [9,17]	12 [9,17]	5 [2,9]	6 [2,9]
Leetspeak	0 [0,0]	0 [0,0]	0 [0,0]	0 [0,0]
Past tense	6 [4,7]	6 [4,7]	5 [3,6]	5 [3,6]
Academic	15 [13,17]	9 [8,12]	4 [2,8]	4 [2,8]
Code	45 [40,51]	14 [10,16]	7 [4,13]	6 [3,13]
Translation	51 [45,55]	3 [2,5]	5 [3,13]	5 [3,8]
Payload split	95 [93,97]	95 [93,97]	95 [93,97]	95 [93,97]
Over-refusal (plain benign)	60.3	60.3	60.6	60.6

Table 6: Judged ASR (%) on half B under realistic traffic, $k_{\max}$=30 (post hoc); mean and 95% interval over 10 windows. Base64 omitted (judge artefact, Section 3). Last row: over-refusal on plain benign requests.

Generation. Table 6 reports judged ASR under realistic traffic. The hybrid rule lowers ASR on fiction from 89% to 37%, on code completion from 45% to 6%, on academic framing from 15% to 4%, and on indirect injection from 12% to 6%. The shared threshold leaves fiction and indirect injection unchanged. Translate-then-answer is already caught by the shared threshold (51% to 3%), and the hybrid rule keeps it there (5%). Over-refusal on plain benign requests rises from 60.3% to 60.6%. The high base rate reflects how often the 1.5B model refuses the borderline prompts of XSTest and OR-Bench even undefended. An always-on steer, by contrast, refuses every plain benign request. The cost of the hybrid rule falls on prefix injection, where ASR rises from 9% (shared) to 16%, and on payload splitting, which stays at 95%. Appendix B gives the pre-registered configuration and a prevalence sweep. With harmful prompts at 20% of every framing the adaptive statistics degrade, and at 35% they fail.

7 Discussion

Activation gates are usually reported by AUROC, and by that measure they generalise to new framings. Deployed gates operate at a threshold, and there they do not. Most of the gap is calibration, and it can be narrowed without new labels by letting the threshold follow the framing. This agrees with observations that probes keep their ranking but lose their operating point off-distribution (McKenzie et al., 2025; Kim and Lee, 2026; Azov et al., 2026). It also differs from settings where the information itself is missing (Schwarz, 2026), which is what we see for base64 on a small model. Defenses that remove framing from the representation, as ARETE attempted, address a change in separability. Here the change was in location.

The remedy's limits follow from its one assumption: each cluster is mostly benign. That holds for dual-use framings such as fiction, retrieved documents, academic or code requests, which are also the hardest to block wholesale. It fails for framings that exist mainly to attack, and those need a fixed floor or a template-level filter. The method also depends on cluster resolution. Over-segmenting is cheap, because small clusters fall back to validation statistics, whereas under-segmenting silently merges framings. We would set $k_{\max}$ well above the expected number of distinct templates.

8 Conclusion

Activation-gated jailbreak defenses fail on withheld framings mainly because their threshold stays

where it was calibrated, while the model still represents the harm. Recalibrating per framing cluster from unlabeled traffic, with the original threshold as a floor, recovers much of the lost protection for dual-use framings at almost no over-refusal cost. It cannot protect framings used mainly for attacks, and it depends on adequate cluster resolution.

Limitations

We study one 1.5B-parameter model. Larger models parse encodings such as base64, which would likely turn that framing from a capability failure into a calibration failure. Each framing uses one to three fixed templates, which makes clustering easier than for paraphrased or attacker-generated framings. The best-performing configuration ($k_{\max}$=30) was chosen after seeing pre-registered results, and the original withheld framings informed the method's design. Only the fresh framings under the pre-registered configuration are fully untouched evidence. We did not test adaptive attackers who optimise against the gate (Bailey et al., 2026), camouflage prompts into benign clusters, or flood a single cluster. The prevalence sweep raises prevalence uniformly and is only a proxy for flooding. ASR is judged on 96-token responses by one model judge, and generation-level ASR is composed from undefended and always-on responses. The traffic mixes are simulated; real prevalence per framing is unknown.

Ethics Statement

This work studies defenses against misuse of language models. Harmful prompts come from public benchmarks, the attack templates are publicly documented, and generated responses are kept out of the released artefacts. We report failure modes, including attack-only framings and cluster under-segmentation, so that defenders do not over-trust activation gates.

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A ARETE Geometry

ARETE standardises activations and maps them through two learned linear heads onto unit hyperspheres for intent and context. It is trained with (i) an invariance loss pulling the intent coordinates of a wrapped prompt towards those of its plain version, (ii) a framing-classification loss on the context sphere, and (iii) a cap loss placing harmful intents inside, and benign intents outside, eight unsupervised spherical caps. The gate score is the maximum cap margin. Steering uses the refusal direction projected onto the orthogonal complement of a rank-16 framing subspace, estimated by SVD of displacements $h_{\text{wrapped}} - h_{\text{plain}}$ on training behaviours.

B Additional Results

Pre-registered configuration at generation level. Table 7 repeats Table 6 with $k_{\max}$=12.

Framing	No defense	Shared threshold	Adaptive (ours)	Hybrid (ours)
Direct	1 [0,2]	0 [0,1]	1 [0,2]	0 [0,1]
Persona	4 [2,6]	1 [0,3]	4 [2,6]	4 [2,5]
Ref. supp.	3 [1,4]	2 [1,3]	3 [1,4]	3 [1,4]
Prefix inj.	76 [70,82]	10 [6,13]	76 [70,82]	17 [13,21]
Fiction	89 [85,92]	87 [83,92]	85 [67,92]	86 [68,93]
Indirect	13 [10,18]	13 [10,18]	13 [10,18]	13 [10,18]
Leetspeak	0 [0,0]	0 [0,0]	0 [0,0]	0 [0,0]
Past tense	6 [4,8]	6 [4,8]	6 [4,8]	6 [4,8]
Academic	15 [11,18]	9 [5,12]	9 [2,16]	9 [3,14]
Code	46 [40,53]	14 [9,18]	11 [2,27]	12 [3,26]
Translation	52 [46,56]	3 [2,5]	5 [0,11]	4 [1,8]
Payload split	95 [93,97]	95 [93,97]	95 [93,97]	95 [93,97]
Over-refusal (plain benign)	59.6	59.6	60.6	60.5

Table 7: As Table 6, pre-registered $k_{\max}$=12, 20 windows.

Prevalence. Table 8 uses the eight original framings, uniform prevalence π and 20 draws.

GCG. Suffixes were optimised for 20 test behaviours against the undefended model (80 steps, 128 candidates per step, top-128 tokens, 20-token suffix). They reach 25% judged ASR undefended. The final-token linear probe lowers this to 10%, the multi-view union to 15%, and ARETE to 22%. With 20 behaviours these numbers are indicative only.

Longer responses. Pending.

		Seen framings (TPR)				Unseen framings (TPR)				
π	Rule	Direct	Persona	Ref. supp.	Prefix inj.	Fiction	Indirect	Base64	Leetspeak	FPR
1%	Shared	98	92	80	92	5	20	0	0	0.5
	Adaptive	88	95	100	92	85	70	0	5	1.8
	Hybrid	98	95	100	92	85	68	0	5	1.7
5%	Shared	95	86	85	91	7	17	0	0	0.5
	Adaptive	80	90	94	88	87	65	0	4	1.6
	Hybrid	93	88	94	87	86	64	0	2	1.5
10%	Shared	96	89	83	92	4	17	0	0	0.5
	Adaptive	73	87	86	82	78	56	0	1	1.3
	Hybrid	94	86	84	82	76	54	0	1	1.3
20%	Shared	97	87	83	92	5	17	0	0	0.5
	Adaptive	28	50	61	56	52	36	0	1	0.8
	Hybrid	95	59	76	81	49	35	0	0	1.0

Table 8: Eight original framings, uniform harmful prevalence π, $k_{\max}$=12; recall (%) and benign FPR (%) averaged over framings. At π=35% (not shown) adaptive recall is at most 5% on every framing.